

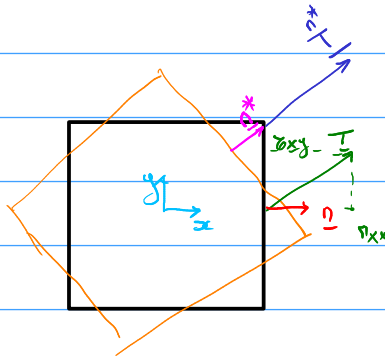
$$\underline{\underline{T}} \rightarrow (x, y)$$

$$\underline{\underline{T}} \rightarrow (x, y) \text{ y } \underline{\underline{n}}$$

Tensiones principales y direcciones principales

$$\underline{\underline{T}} = \underline{\underline{\sigma}} \cdot \underline{\underline{n}} = \lambda \underline{\underline{n}}$$

Búsqueda
 $\underline{\underline{T}} \parallel \underline{\underline{n}}$



$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix}$$

$$\underline{\underline{\sigma}} \cdot \underline{\underline{n}} - \lambda \underline{\underline{n}} = 0$$

$$(\underline{\underline{\sigma}} - \lambda \underline{\underline{I}}) \cdot \underline{\underline{n}} = 0 \rightarrow \text{sistema de ecuaciones homogéneas}$$

$$\det(\underline{\underline{\sigma}} - \lambda \underline{\underline{I}}) = 0$$

→ Ecuación característica → $\lambda^{(i)}$ autovalores

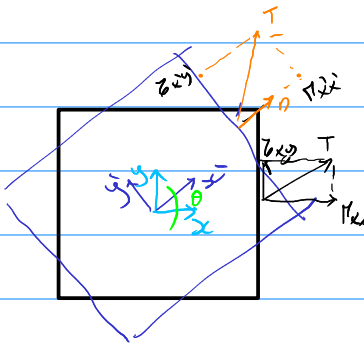
*

$$\sigma'_{ij} = \beta_{im} \beta_{jn} \sigma_{mn}$$

$$\underline{\underline{\sigma'}} = \underline{\underline{\beta}} \underline{\underline{\sigma}} \underline{\underline{\beta}}^T$$

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix}$$

$$\underline{\underline{\sigma'}} = \begin{bmatrix} \sigma'_{xx} & \sigma'_{xy} \\ \sigma'_{xy} & \sigma'_{yy} \end{bmatrix}$$



$$\frac{\partial \sigma'_{xx}}{\partial \theta} = 0$$

$$(*) \quad \sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta, \quad (4.2-6)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta, \quad (4.2-7)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta. \quad (4.2-8)$$

From these equations, it follows that

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y, \quad (4.2-9)$$

$$\frac{\partial \sigma_{x'}}{\partial \theta} = 2\tau_{x'y'}, \quad \frac{\partial \sigma_{y'}}{\partial \theta} = -2\tau_{x'y'}, \quad (4.2-10)$$

$$\tau_{x'y'} = 0 \text{ when } \tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \quad (4.2-11)$$

$$\sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$\sigma_{\min}$

radio del círculo

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

direcciones principales

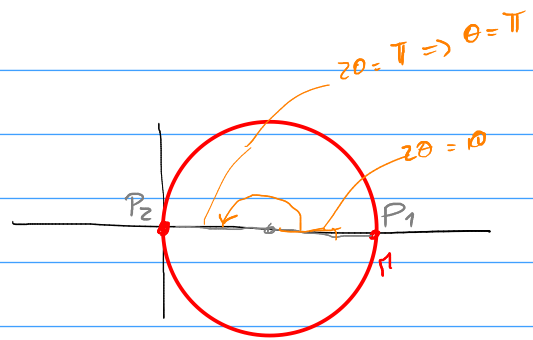
7) a) Tensión/tracción uniaxial pura

$$\sigma = \begin{bmatrix} \sigma & 0 \\ 0 & 0 \end{bmatrix}$$



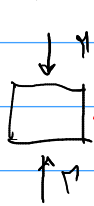
$$P_1 = (\sigma, 0)$$

$$P_2 = (0, 0)$$



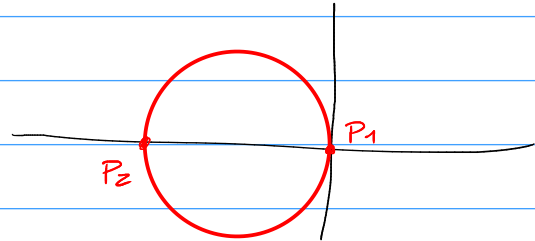
b) Compresión uniaxial pura

$$\sigma = \begin{bmatrix} 0 & 0 \\ 0 & -\sigma \end{bmatrix}$$



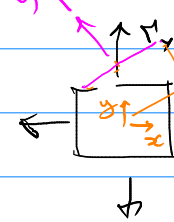
$$P_2 = (-\sigma, 0)$$

$$P_1 = (0, 0)$$



c) Torsión biaxial pura

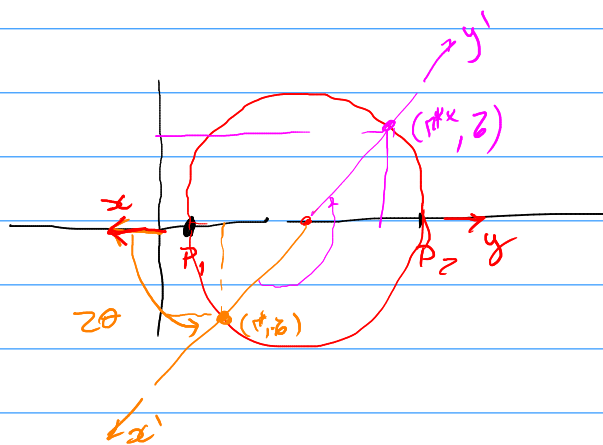
$$\sigma = \begin{bmatrix} \tau_x & 0 \\ 0 & \tau_y \end{bmatrix}$$



$$P_2 = (\tau_y, 0)$$

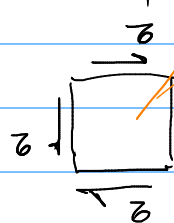
$$P_1 = (\tau_x, 0)$$

$$\tau_x < \tau_y$$



d) Corte pura

$$\sigma = \begin{bmatrix} 0 & \tau \\ \tau & 0 \end{bmatrix}$$

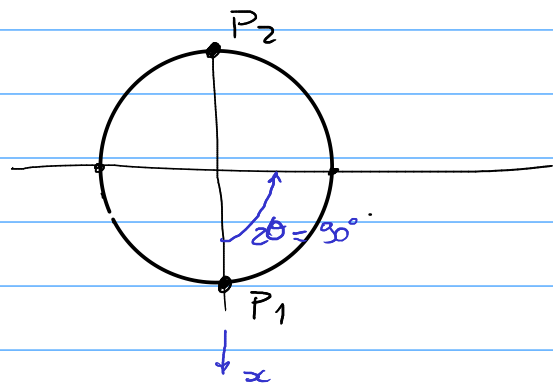


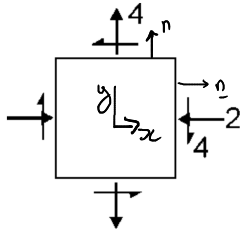
$$P_2 = (0, \tau)$$

$$\tau > 0$$

$$\theta = 45^\circ$$

$$P_1 = (0, -\tau)$$





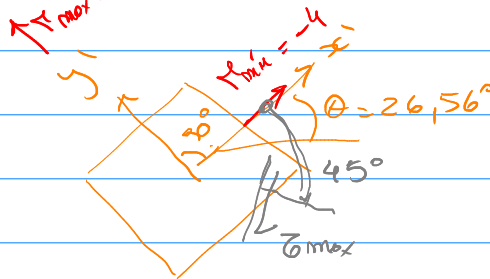
$$\sigma = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix} = \begin{bmatrix} -2 & -4 \\ -4 & 4 \end{bmatrix}$$

$$\sigma_{\max/\min} = \frac{-2+4}{2} \pm \sqrt{\left(\frac{-2-4}{2}\right)^2 + (-4)^2} = 1 \pm 5 \rightarrow \begin{matrix} \sigma_{\max} = 6 \\ \sigma_{\min} = -4 \end{matrix}$$

$$\tan(2\theta) = \frac{2 \cdot -4}{-2 - 4} = \frac{-8}{-6} = \frac{4}{3} \Rightarrow \theta = 26,56^\circ$$

a(θ) para determinar si corresponde a σ_{max} o σ_{min}

$$\theta = 26,56^\circ + 90^\circ = 116,56^\circ$$



$$\sigma_{\max} = \frac{6 - (-4)}{2} = 5$$

• Círculo de Mohr

$$P_1 = (-2, 4)$$

$$P_2 = (4, -4)$$

